06. Symmetry

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Lines of symmetry of a regular shape:

How many lines of symmetry does a regular octagon have?

Answer: A regular octagon has 8 equal sides, and a regular shape always has the same number of lines of symmetry as it has sides. Answer: **8**.

Example 2 — Reflecting a point in the x-axis:

Reflect the point (4, 9) in the x-axis (the line y = 0). Give the coordinates of the image.

Answer: Reflecting in the x-axis keeps the x-coordinate the same and changes the sign of the y-coordinate: the image is **(4, -9)**.

Example 3 — Reflecting a point in the y-axis:

Reflect the point (-6, 3) in the y-axis (the line x = 0). Give the coordinates of the image.

Answer: Reflecting in the y-axis keeps the y-coordinate the same and changes the sign of the x-coordinate: the image is **(6, 3)**.

Example 4 — Order of rotational symmetry:

What is the order of rotational symmetry of a regular pentagon?

Answer: A regular pentagon looks exactly the same 5 times during one full 360° turn. Its order of rotational symmetry is **5**.

Example 5 — Multi-step symmetry and area:

A shape has one horizontal line of symmetry. The top half has an area of 24 cm². What is the total area of the shape?

Answer: A line of symmetry splits a shape into two identical (mirror-image) halves, so the bottom half must also be 24 cm². Total area = 24 + 24 = **48 cm²**.

Foundation

15 questions

1. How many lines of symmetry does a square have?

2. How many lines of symmetry does a regular pentagon have?

3. How many lines of symmetry does an equilateral triangle have? A) 1 B) 2 C) 3 D) 4

4. How many lines of symmetry does a rectangle (that is not a square) have?

5. How many lines of symmetry does the capital letter A have?

6. How many lines of symmetry does the capital letter E have?

7. Which capital letter has exactly 2 lines of symmetry? A) P B) X C) R D) F

8. How many lines of symmetry does a regular hexagon have?

9. How many lines of symmetry does an isosceles triangle have (a triangle with exactly two equal sides, not equilateral)?

10. Select ALL of these shapes that have at least one line of symmetry: A) Square B) Scalene triangle C) Isosceles triangle D) Regular pentagon E) Parallelogram (not a rectangle or rhombus)

11. Reflect the point (3, 8) in the x-axis. Give the coordinates of the image.

12. Reflect the point (-5, 2) in the y-axis. Give the coordinates of the image.

13. How many lines of symmetry does a rhombus have (assuming it is not a square)?

14. How many lines of symmetry does a regular octagon have?

15. A kite shape has how many lines of symmetry?

Standard

20 questions

1. What is the order of rotational symmetry of a square?

2. What is the order of rotational symmetry of an equilateral triangle?

3. What is the order of rotational symmetry of a regular hexagon? A) 2 B) 3 C) 4 D) 6

4. What is the order of rotational symmetry of a parallelogram that is not a rectangle?

5. A regular heptagon has 7 sides. What is its order of rotational symmetry, and how many lines of symmetry does it have?

6. Which of these shapes has rotational symmetry of order 2 but NO lines of symmetry? A) Rectangle B) Square C) Parallelogram (not a rectangle or rhombus) D) Isosceles trapezium

7. Reflect the point (6, 1) in the line y = x. Give the coordinates of the image.

8. Reflect the point (-4, 7) in the x-axis. Give the coordinates of the image.

9. Select ALL of these capital letters that have rotational symmetry of order 2: A) H B) N C) F D) S E) Z

10. Reflect the point (2, -3) in the y-axis. Give the coordinates of the image.

11. A shape has a horizontal line of symmetry. The bottom half has an area of 15 cm^2. What is the total area of the shape?

12. How many lines of symmetry does a regular nonagon (9 sides) have?

13. Which of these letters has exactly 2 lines of symmetry? A) X B) P C) F D) J

14. Reflect the point (0, 5) in the x-axis. Give the coordinates of the image.

15. A square has a vertex at (2, 6). The square is reflected in the line y = x. What are the new coordinates of that vertex?

16. What is the order of rotational symmetry of a regular octagon?

17. Select ALL of the following statements that are TRUE: A) A circle has infinite lines of symmetry B) A square has rotational symmetry of order 4 C) A scalene triangle has no lines of symmetry D) A rectangle (not a square) has 4 lines of symmetry

18. Reflect the point (-7, -2) in the y-axis. Give the coordinates of the image.

19. How many lines of symmetry does a kite have?

20. A regular polygon has 10 sides. How many lines of symmetry does it have, and what is its order of rotational symmetry?

Hard

5 questions

1. A shape has two lines of symmetry: one horizontal and one vertical. One quarter of the shape (the top-right section) has an area of 12 cm^2. What is the total area of the whole shape?

2. Point P is at (3, 4). It is reflected in the x-axis to give point Q, then Q is reflected in the y-axis to give point R. What are the coordinates of R?

3. A regular polygon has rotational symmetry of order 12. How many sides does it have, and how many lines of symmetry does it have?

4. Select ALL the statements below that are TRUE: A) Every regular polygon has the same number of lines of symmetry as it has sides B) A parallelogram (not a rectangle or rhombus) has 2 lines of symmetry C) The order of rotational symmetry of any shape is always at least 1 D) A rectangle that is not a square has rotational symmetry of order 2

5. Point (5, -2) is reflected in the line y = x to give point A, then point A is reflected in the x-axis to give point B. What are the coordinates of B?

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